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RESEARCH ARTICLE

Toward Generalization of Deep Learning-Based Plant Disease Identification Under Controlled and Field Conditions

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ABSTRACT Identifying corn diseases under field conditions is crucial for implementing effective disease management systems. Deep learning (DL)-based plant disease identification using deep neural networks (DNN) has been successfully implemented in recent years. Recent work suggests DL models trained on lab-acquired image data do not generalize to similar accuracy levels for identifying diseases in the field. Additionally, most studies have not evaluated the generalizability of DL models for identifying plant diseases from various datasets and diverse imaging conditions. This study evaluates how well DL models generalize across different datasets and environmental conditions for identifying plant diseases using five datasets consisting of foliar disease images in corn: namely PlantVillage, PlantDoc, Digipathos, northern leaf blight (NLB) dataset, and a custom acquired CD&S dataset. Multiple DL-based image classification models were trained and evaluated using different dataset combinations. Transfer learning was utilized using five different pre-trained DNN architectures: InceptionV3, ResNet50, VGG16, DesneNet169, and Xception, for conducting four different experiments. After training the models, images for distinct corn diseases from different datasets were used as testing data for evaluating the generalization ability of each DL model. It was observed that the DenseNet169 model performed the best. The highest generalization accuracy of 81.60% was observed when the DenseNet169 model was trained using red, green, blue, and alpha (RGBA) images from CD&S corn disease dataset with removed backgrounds. In addition, 77.50% to 80.33% accuracy was achieved for the PlantVillage dataset when combined with field-acquired images from either the PlantDoc or the CD&S dataset. The results suggest that background removal using RGBA images from CD&S dataset or augmentation of field and lab data improves the generalization performance of DL models and could be used for developing field-deployable disease management systems.

INDEX TERMS Deep learning, disease identification, generalization, image classification.

I. INTRODUCTION

Diseases in crops and plants cause significant yield losses resulting in high disease management costs and profit loss for farmers worldwide. Corn is a globally important crop, accounting for 95% of feed crops in the US [1]. It suffers 2% to 15% yield losses due to diseases [2]. To implement an effective disease management system, it is important to first identify diseases under various different conditions accurately. Traditional disease management approaches required manual scouting of fields resulting in delayed crop disease analysis. In addition to being inefficient, traditional methods are also subjective due to visual analysis by expert plant pathologists [3]. To overcome the limitations presented by traditional crop disease analysis, fast and accurate disease identification approaches were developed using computer

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vision-based image analysis. However, most solutions relied on feature engineering to define the required features corresponding to each unique disease. As various plants, crops, and their respective diseases exist, each would require a certain set of features to be extracted for training computers to identify them accurately. Therefore, reliance on deep learning (DL) has recently increased to develop generalizable solutions. DL has gained popularity within the past decade due to technological advancements in graphical processing units (GPUs), memory capacity, and dataset availability. Modern approaches to tackle plant diseases include using deep neural networks (DNNs) to automatically obtain features from imagery training data that are important for training disease identification models. This approach helps bypass feature extraction and feature engineering requirements when developing disease management systems.

For plant disease identification, DL models generally rely on high-level features that can be extracted from visual symptoms. Therefore, access to good imagery datasets is essential. Ever since the PlantVillage dataset was introduced, DL model-based solutions have become popular for plant disease identification [4]. Additionally, new datasets are constantly being made available to the public, such as the Digipathos [5], PlantDoc [6], and Northern Leaf Blight (NLB) datasets [7].

The PlantVillage dataset was used to identify all the 38 classes corresponding to 26 diseases and 12 healthy classes within the dataset with testing accuracies of up to 95.81% [8], 97.82% [9], 98.3% [10], and 99.35% [11]. Support vector machine (SVM) and K Nearest Neighbors (KNN) are traditional machine learning algorithms that were trained by extracting features from PlantVillage using DNN to identify multiple diseases with up to 97.82% accuracy. In addition, the study proved that DL-based solutions performed better than traditional machine learning-based solutions developed using manual feature extraction [9]. Diseases corresponding to individual crops from PlantVillage were also identified with testing accuracies up to 94.3% for grape disease identification [12] and 99.18% for tomato disease identification [13]. Another large plant disease dataset that was commonly used within the literature for DL model training was the Digipathos dataset. Soybean-related diseases were identified with up to 93.03% testing accuracy after the Digipathos dataset was used for training DL models [14]. Additionally, the PlantVillage and Digipathos datasets were used to identify diseased leaves and lesions with testing accuracies of 99.93% and 99.74%, respectively [15]. PlantVillage and Digipathos datasets were also used simultaneously in additional studies [16], [17]. The PlantDoc dataset is a much smaller dataset. Due to its small size, its applications within the research community were limited. PlantDoc was used for disease identification [18], [19]. Most datasets that have been used within the literature were used for the identification of diseases in multiple crops. Some studies, however, have focused on identifying single diseases. A custom unmanned aerial system (UAS) and handheld imagery dataset of NLB-infected corn [7] was used to identify NLB lesions with millimeter accuracy [20]. The NLB dataset was also used for identifying NLB disease lesions by additional studies [21], [22], [23]. The overall results reported in the literature are shown in table 1.

TABLE 1. Summary of results reported in literature.

Study	Dataset	Testing Accuracy
[8]	PlantVillage	95.81%
[9]	PlantVillage	97.82%
[10]	PlantVillage	98.3%
[11]	PlantVillage	99.35%
[25]	PlantVillage	99.53%
[26]	PlantVillage	99.01%
[15]	PlantVillage	99.93%
[16]	PlantVillage	94.52%
[27]	PlantVillage	99.84%
[12]	PlantVillage (Grape)	94.3%
[13]	PlantVillage (Tomato)	99.18%
[28]	PlantVillage (Tomato)	99.51%
[14]	Digipathos (Soybean)	93.03%
[15]	Digipathos	99.74%
[16]	Digipathos	86.38%
[17]	PlantVillage & Digipathos	98.7%
[6]	PlantDoc	70.53%
[18]	PlantDoc	69.3% (mAP)
[19]	PlantDoc	62.76%
[20]	NLB Dataset	99.79%
[23]	NLB Dataset	95.1%
[21]	NLB Dataset	96.7%
[22]	NLB Dataset	93%
[29]	CD&S	98.34%

One major limitation in current disease identification solutions that rely on DL is the inability of trained models to generalize to data from different datasets and conditions. Datasets such as PlantVillage and Digipathos are limited in their ability to train robust models due to lack of variations in lighting, size, or framing [24]. Although high testing accuracies of over 99% were reported for models trained using the PlantVillage dataset, the testing accuracy dropped to below 50% when the images from field conditions were provided to DL models trained on the PlantVillage dataset [25], [26]. When hidden testing images corresponding to different conditions were used from the same dataset, testing accuracies of 99.53% [25] and 99.01% [26] were reported. The testing accuracies dropped to as low as 33.27% and 33.97% after field-acquired images were used for testing models trained on lab-acquired images. However, an improved accuracy of 65.69% was observed when lab-acquired images were used for testing models trained on field-acquired images. The original study, which introduced the PlantDoc dataset, also reported the generalization of DL by training using PlantVillage and testing using the PlantDoc dataset with field conditions, which resulted in testing accuracies of up to 39.87%. In addition, PlantVillage dataset images with uniform backgrounds were recently used for DL-based disease identification in fields. The generalization performance of the models improved from 39.38% to 72.03% [27]. Generalization results from the literature are shown in table 2.

In this study, a total of four experiments were conducted. DL-based image classification models were first trained

TABLE 2. Generalization from controlled to field conditions.

Study	Train / Test (Dataset Type)	Testing Accuracy (Field)
[25]	Lab (PlantVillage) / Lab (PlantVillage)	99.53%
	Lab (PlantVillage) / Field	33.27%
	Field / Lab (PlantVillage)	65.69%
[26]	Lab (PlantVillage) / Lab (PlantVillage)	99.01%
	Lab (PlantVillage) / Field (IPM)	45.95%
	Lab (PlantVillage) / Field (Bing)	33.97%
[6]	Lab (PlantVillage) / Field (PlantDoc)	39.87%
[27]	Lab (PlantVillage) / Lab (PlantVillage)	**99.84%**
	Lab (PlantVillage) / Field	39.38%
	Lab (PlantVillage) / Field	**72.03%**

using five DNN architectures, namely VGG16, ResNet50, InceptionV3, DenseNet169, and Xception. Each of the five architectures was used to train models using the PlantVillage dataset, PlantDoc dataset, CD&S dataset, and red, green, blue, and alpha (RGBA) CD&S dataset without background. Models were then trained using only the images corresponding to the NLB and Gray Leaf Spot (GLS) diseases. The same five architectures were also used to train models by combining different dataset types for generalization. Once all the models were trained, the generalization capability of the models was evaluated using testing data across datasets with diverse imaging conditions.

The remaining study is organized by first introducing the datasets, DL techniques, and experiments in the ''Methodology'' section. Results and analysis are then presented in the ''Experimental Results'' section. The results are then further discussed and compared with the literature within the ''Discussion'' section. Finally, the conclusions are given in the ''Conclusions'' section.

II. METHODOLOGY

A. DATASETS

Five different datasets were used in this study for testing DL model generalization across datasets for plant disease identification (figure 1). Four of the datasets, namely PlantVillage, PlantDoc, Digipathos, and the NLB dataset [16], were publicly available. Additionally, three versions of a custom dataset acquired in field conditions were also used. The information for datasets was summarized in table 3.

1) PLANTVILLAGE DATASET

PlantVillage [4] is the largest publicly available dataset for plant disease identification. As this study was focused on identifying corn diseases, only the four corn disease classes from PlantVillage were used: namely healthy, NLB, GLS, and Common Rust disease classes with 2324, 2385, 2052, and 2384 lab-acquired images, respectively.

2) PLANTDOC DATASET

The PlantDoc dataset [6] is another publicly available dataset with similar classes and diseases to the PlantVillage. However, the PlantDoc dataset is significantly smaller as it

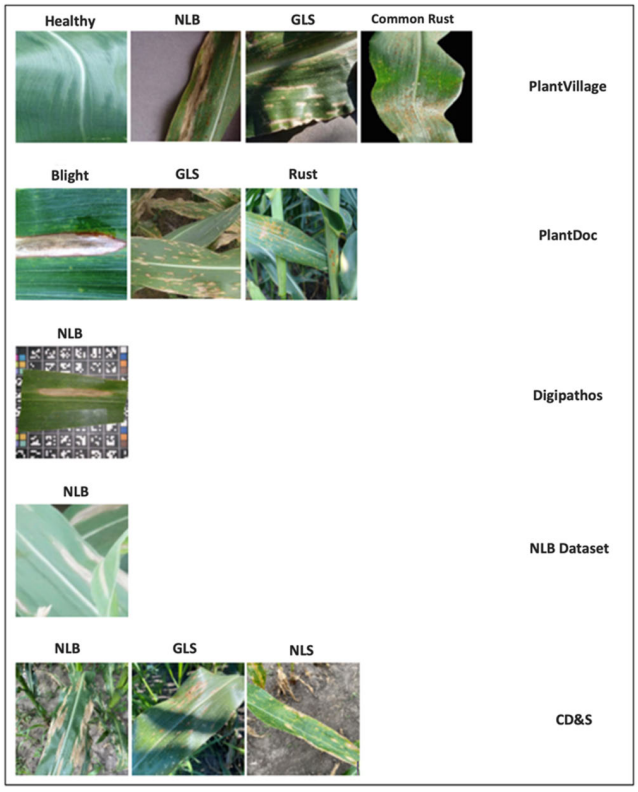


FIGURE 1. Sample images from the PlantVillage, PlantDoc, Digipathos, NLB, and CD&S datasets used in this study.

TABLE 3. Summary of datasets used in this study.

Dataset	Class	Images
PlantVillage	Northern Leaf Blight	2324
	Gray Leaf Spot	2385
	Common Rust	2052
	Healthy	2384
PlantDoc	Northern Leaf Blight	191
	Gray Leaf Spot	68
	Rust	116
Digipathos	Northern Leaf Blight	202
NLB Dataset	Northern Leaf Blight	105
CD&S Dataset	Northern Leaf Bligh	497
	Gray Leaf Spot	523
	Northern Leaf Spot	551
Total Images		11,398

contains a total of 2598 images. The images within the PlantDoc dataset consist of corn diseases acquired in field conditions, however, most images were acquired from online sources resulting in a very variable dataset. Only the corn disease images were used for this study, including the blight, GLS, and rust classes, with 191, 68, and 116 images. The blight class was renamed NLB for this study and NLB images were carefully chosen. Additional data preparation included removing images that would not help DL models learn relevant disease features.

3) DIGIPATHOS DATASET

Another large publicly available dataset, Digipathos [5], consists of multiple plant disease images across multiple crops

and diseases. However, this dataset consists of different corn disease classes with only the NLB class, which was common to the other datasets used in this study. Therefore, the 202 NLB images from this dataset were used for testing across datasets, of which 66 leaf images were chosen.

4) NORTHERN LEAF BLIGHT (NLB) DATASET

Finally, the NLB dataset [7] acquired in field conditions using a handheld sensor was used for testing purposes. Only 105 images, consisting of NLB lesions, were carefully selected using visual observation to identify NLB in field conditions using DL.

5) CD&S DATASET

Corn Disease & Severity (CD&S), a custom dataset acquired at the Purdue Agronomy Center for Research and Education (ACRE), consisting of diseases common to corn production systems in Indiana, US, was also used [30]. CD&S consists of 497, 523, and 551 images of NLB, GLS, and Northern Leaf Spot (NLS) diseases from the field, respectively. Additionally, the removed background version of the CD&S dataset with RGBA images was also utilized for training DL models. The background of RGB images from CD&S dataset was removed/made transparent by introducing an alpha (A) channel to RGB image and setting its value to zero. Further, the pixel values corresponding to Red, Green, and Blue (RGB) channels were randomized using Python. For evaluating generalization performance, two modified versions of the CD&S dataset with uniform white and black backgrounds were also utilized. As 50% of the data was used for training purposes, each dataset version consisted of 781 images.

B. DEEP LEARNING

1) CONVOLUTIONAL NEURAL NETWORKS

DL is a technique of machine learning that uses DNNs. Unlike conventional image processing and machine learning techniques that rely on manual feature extraction, DL can automatically extract important features from images for training models. Many studies rely on creating custom DNN architectures. However, there exist some popular DNN architectures that are commonly used. DL models require large datasets and powerful computational resources such as GPUs. Furthermore, DL model training can be time-consuming. Therefore, many studies with limited dataset sizes and computational resources rely on transfer learning. Transfer learning is an approach for training DL models using pre-trained weights from different trained models. The most common pre-trained weights are derived from models trained on the ImageNet dataset [4]. It has been reported that using transfer learning for training DL models on small datasets can result in higher accuracies and faster training times than training models from scratch [31].

Multiple different network architectures, datasets, and dataset combinations were used in this study to train and assess the generalizability of different DL models to identify plant diseases. The InceptionV3 [32], ResNet50 [33], VGG16 [34], DenseNet169 [35], and Xception [36] were used.

InceptionV3 is a popular DNN architecture that builds on the GoogLeNet architecture introduced by Google. InceptionV3 was built using various techniques to improve the model's overall computational and memory efficiency. The techniques used include factorizing convolutions, which essentially reduce parameters and thus enhance the overall efficiency of the model. Asymmetric convolutions were used to reduce the parameters further. Smaller convolutions were also used to increase the training speed. Additionally, an auxiliary classifier was used as a regularization technique. Overall, the InceptionV3 architecture was also more efficient than the VGGNet family of architectures.

ResNet50 is another DNN architecture that consists of ResNet blocks with skip connections. ResNet uses residual mappings to overcome the problems associated with degradation resulting from saturation as DL models become deeper. ResNet50 was then developed along with additional versions, including ResNet18, ResNet34, ResNet101, and ResNet152, with varying numbers of residual layers. ResNet50, however, is commonly used. The layers within the ResNet architectures rely on skip connections. The ResNet architecture is less complex than the VGG16 architecture as there are fewer filters and fewer number of overall parameters.

VGG16 is an older DNN architecture with fewer layers than the other networks, however, VGG16 has significantly more trainable parameters that increase the complexity of the overall network. Therefore, training times are longer and require greater memory. Although VGG16 was introduced very early in the rise of DL, it is a common architecture that is used for comparisons and benchmarking purposes. In addition, VGG16 resulted in the highest testing accuracies for disease region identification, disease type classification, and disease severity estimation for corn disease diagnosis [29].

DenseNet169 belongs to the DenseNet DNN architecture family that also relies on skip connection. However, unlike ResNet50, where multiple different ResNet blocks have skip connections within each block, DenseNet169 consists of five different dense blocks connected to every other following dense block. This structure allows the model to retain high-level features that are commonly learned at the start of the architecture. For the purposes of plant disease identification, DenseNet can prove to be of great use as high-level features corresponding to the color, shape, and texture of the disease lesions are necessary for training disease identification models.

Finally, Xception (extreme inception) is another popular DNN architecture that relies on the Inception architectures; however, the Inception modules are replaced by depthwise separable convolution. Essentially, 1×1 convolutions are used to compress the input. In addition, unlike

in Inception architectures, Xception does not introduce non-linearity. Xception is one of the more recent developments in DNN architectures and has been used to report promising results in various applications.

2) HYPERPARAMETERS

This study used ImageNet pre-trained weights to train five different models for each experiment. The InceptionV3, ResNet50, VGG16, DenseNet169, and Xception were used. When training the models, 35% - 15% - 50% training - validation - testing split was used to ensure that the testing accuracy was obtained from half the dataset. Each image was resized to 299 $\times$ 299 pixels for the InceptionV3 model and 224 $\times$ 224 pixels for all others, as specified by the network architecture recommendations. Using a batch size of 16, each model was trained for 100 epochs. In addition, the Adam optimizer with an initial learning rate of 0.0001 and the categorical cross-entropy loss function was used.

3) DATASET AUGMENTATION

Dataset augmentation is a technique that is generally used to expand datasets and increase the diversity of data to help enhance DL model performance under different conditions. In this study, the Keras built-in dataset augmentation techniques were used to train each DL model using the ImageDataGenerator function. For augmentation, images were flipped horizontally, rotated by 40 degrees, subjected to a zoom factor of 0.2, and their brightness changed within the range of 0.2 to 1.0. Although dataset augmentation by modifying training images can help improve DL model performance, it is crucial to test their performance for plant disease identification when generalizing to different conditions. Generalization across datasets and data acquisition conditions are further explained in section C under ''Generalization Across Datasets''.

4) EVALUATION METRICS

After training all the different models for assessing the generalization performance of the models, the testing accuracies were calculated for each class using equation 1.

$$testing\ accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

where TP, FP, TN, and FN are the true positive, false positive, true negative, and false negative values, respectively. After obtaining the testing accuracy corresponding to each dataset class, the testing accuracies were averaged. The average testing accuracies were then compared to assess the generalization performance of each model trained using different datasets.

5) EQUIPMENT

Python's TensorFlow DL framework was utilized to prepare the DL models. Each model was trained using the NVIDIA Tesla V100 GPU with 16 cores available through the Purdue University Gilbreth Community Cluster.

C. EXPERIMENTS

1) DIFFERENT DATASET DISEASE IDENTIFICATION

The first experiment was conducted to identify corn diseases in publicly available and custom-created CD&S datasets. Each of the five DNN architectures, namely InceptionV3, ResNet50, VGG16, DenseNet169, and Xception, was used to train models using corn disease datasets, namely PlantVillage, PlantDoc, CD&S, and CD&S dataset with removed background, resulting in a total of 20 trained models. The models were trained by keeping the training-validation-testing split and hyperparameters consistent across the datasets. Training and validation accuracy were visualized to compare the performances of different models. Hidden testing data from the same dataset was then used to calculate testing accuracies. After comparing the testing accuracies for each model, the best-performing model was identified and reported.

2) BINARY NLB & GLS DISEASE IDENTIFICATION

The same DNN architectures were then used to train models using only the NLB and GLS images from the PlantVillage, PlantDoc, and CD&S datasets. These two diseases were common to most datasets and represented two of the most common corn diseases. Furthermore, by training on only these diseases, the model tried to learn specific NLB and GLS features rather than other diseases which may not be present for generalization assessment. Testing accuracies were also compared to identify the best-performing model.

3) COMBINING UNIFORM & FIELD CONDITION IMAGES

Dataset augmentation by modifications made to the images within the training data is an effective way to help improve generalization performance. Therefore, DL models were also trained using augmented datasets by combining images from different acquisition conditions and datasets to evaluate and improve generalization. Only the NLB and GLS disease images were used for dataset combination as these diseases were common to most of the datasets used in this study. Three dataset combinations were created for training the same five models: the PlantVillage dataset was combined with the PlantDoc dataset, PlantVillage was combined with the CD&S dataset, and finally, the PlantVillage, PlantDoc, and CD&S datasets were combined. This experiment used the same five DNN architectures to train models for each dataset combination. Testing accuracy was then reported using hidden testing data from PlantVillage, PlantDoc, and CD&S datasets.

4) GENERALIZATION ACROSS DATASETS

In this study, generalization refers to the potential of DL models trained using imagery data acquired from one dataset and then tested on data from different datasets or image acquisition conditions. DL model generalization across datasets and acquisition conditions remains a gap in the literature [37], [38], [39]. The generalization of DL models is illustrated

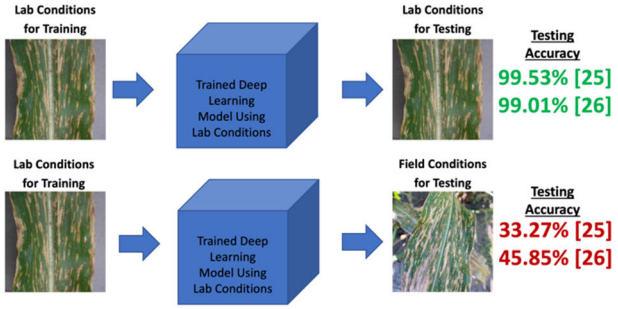


FIGURE 2. Examples of generalization of DL models as shown in studies conducted by [25] and [26].

in figure 2. Therefore, in this study, after each model was trained and compared, testing data from different datasets, acquisition conditions, and backgrounds were provided to each model from the previous experiments to assess how well DL models generalize to hidden testing data from multiple testing datasets. As each dataset consists of a different distribution of corn diseases, only part of the data was used for this experiment. The accuracies obtained for generalizing DL models across different datasets and dataset acquisition conditions, corresponding to both lab and field conditions with controlled and complex noisy backgrounds, respectively, were reported and compared to the generalizations reported in the literature.

III. EXPERIMENTAL RESULTS

A. DIFFERENT DATASET DISEASE IDENTIFICATION

1) PLANTVILLAGE DATASET

After using multiple different network architectures for training the different models, their performances were compared by plotting the training and validation accuracies and losses, as shown in figure 3. In addition, testing accuracies were compared by first using the hidden testing images from the same dataset used for training. Furthermore, testing was also conducted to assess the generalization of the trained DL models.

Each model trained using the PlantVillage dataset reached training and validation accuracies up to 100%. The slight difference between training and validation accuracy indicated a low overfitting degree. Therefore, the PlantVillage dataset is a good dataset that can be used to train image classification models to identify corn diseases accurately. However, as all the training images are acquired under controlled conditions, the model will likely fail under field conditions. The plots for training and validation accuracy and loss can be seen in figure 3.

For further evaluating DL model performance, testing images hidden during training were fed into the model to calculate the testing accuracy, as shown in table 4. The testing accuracies indicate that each of the DL models could identify corn diseases with high accuracies and DenseNet169 achieved the highest testing accuracy of 99.50%.

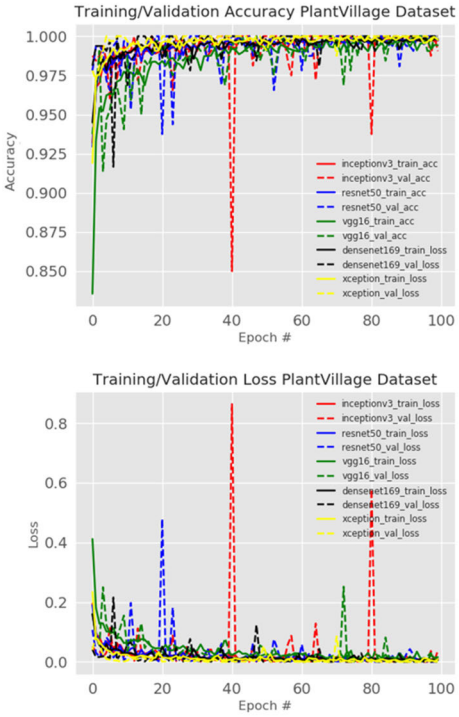


FIGURE 3. Training and validation of DL models using the PlantVillage dataset.

TABLE 4. Results for plantVillage dataset.

Model	Healthy	NLB	GLS	Common Rust	Total
InceptionV3	98.87	95.86	95.30	99.83	97.47
ResNet50	100	99.11	96.22	99.83	98.79
VGG16	100	97.76	97.82	100	98.90
DenseNet169	100	99.62	98.55	99.83	**99.50**
Xception	99.74	98.43	98.74	99.75	99.17

All values are in percentages.

2) PLANTDOC DATASET

Access to datasets similar to PlantVillage, but with field conditions is important. Therefore, images of corn diseases from the PlantDoc dataset were used for training the same five models. Training and validation plots can be seen in figure 4. Greater differences in training and validation curves were observed during model training compared to training using the PlantVillage dataset. This can be attributed to the limited number of images for each disease, the unbalanced dataset, and the field conditions for PlantDoc.

Furthermore, after visual inspection, it was observed that many images within the PlantDoc dataset had been acquired from different online sources with watermarks. There also exist images with entire crops rather than images of leaves. These variations within the dataset did not allow the models to train well, resulting in overfitting.

Testing accuracies were again obtained and reported for each disease and model (table 5). The Xception model achieved the highest testing accuracy of 81.53%. It is clear that although the PlantDoc dataset mimics field conditions,

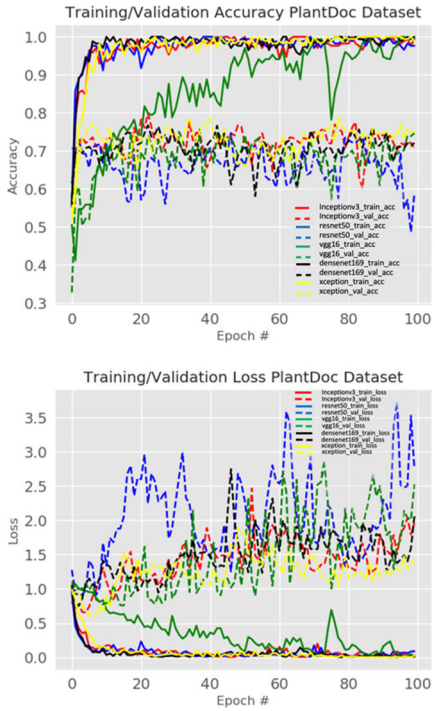


FIGURE 4. Training and validation of DL models using the PlantDoc dataset.

TABLE 5. Results for PlantDoc dataset.

Model	NLB	GLS	Common Rust	Total
InceptionV3	94.83	56.82	78.18	78.34
ResNet50	62.07	68.18	23.64	50.32
VGG16	93.10	72.73	52.27	74.52
DenseNet169	96.55	59.09	65.45	75.15
Xception	91.38	61.38	87.27	**81.53**

All values are in percentages.

TABLE 6. Results for CD&S dataset.

Model	NLB	GLS	NLS	Total
InceptionV3	100	98.82	98.91	99.25
ResNet50	99.19	99.23	99.28	99.23
VGG16	98.79	99.23	97.83	98.62
DenseNet169	99.60	99.62	99.64	**99.62**
Xception	99.60	99.62	98.55	99.26

All values are in percentages.

the dataset is not well constructed, which results in a high degree of overfitting and low testing accuracies.

3) CD&S DATASET

In addition to the publicly available PlantVillage and PlantDoc datasets, the custom CD&S dataset was also used. The five models were first trained to identify corn's NLB, GLS, and NLS diseases in complex field conditions (figure 5).

After training was complete, testing accuracies for each model were obtained. The testing accuracies can be seen in table 6 and DenseNet169 achieved the highest testing accuracy of 99.62%. These results indicate that DL can identify

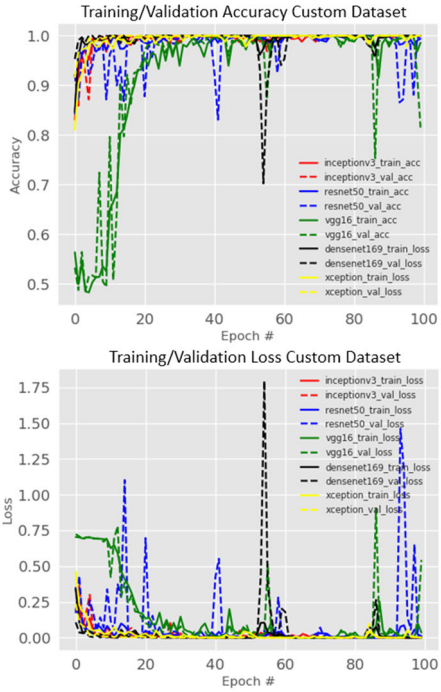


FIGURE 5. Training and validation of DL models using the custom CD&S dataset.

TABLE 7. Results for CD&S dataset with removed background.

Model	NLB	GLS	NLS	Total
InceptionV3	98.39	57.09	89.86	81.66
ResNet50	94.35	82.38	98.91	91.97
VGG16	0	0	100	33.33
DenseNet169	98.39	68.20	96.01	**87.52**
Xception	87.10	83.52	88.77	86.47

All values are in percentages.

diseases in corn within field conditions with high accuracies of over 99%. Furthermore, the CD&S dataset is more balanced with each image containing foliar diseases.

4) CD&S DATASET WITH REMOVED BACKGROUND

DL models achieved high accuracies for corn disease identification using images acquired under field conditions. For further assessing DL model performance, the backgrounds were removed from the CD&S dataset, and the models were trained again. The results can be seen in the accuracy and loss plots in figure 6.

The testing accuracies were reported after the models were trained, as shown in table 7. For this experiment, the testing images did not consist of removed backgrounds. The testing images for this experiment were obtained from the original CD&S dataset with the field background conditions. The DenseNet169 model achieved the highest testing accuracy of 87.52%.

B. BINARY NLB & GLS IDENTIFICATION

1) PLANTVILLAGE DATASET

After evaluating the models for identifying different corn diseases in different datasets, it was observed that the NLB

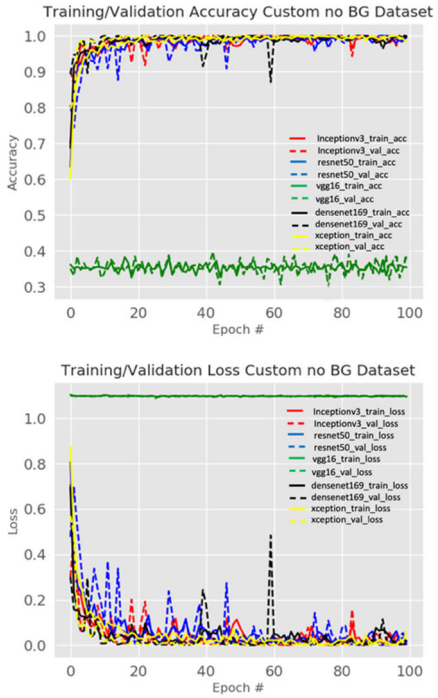


FIGURE 6. Training and validation of DL models using the CD&S dataset after removing the background.

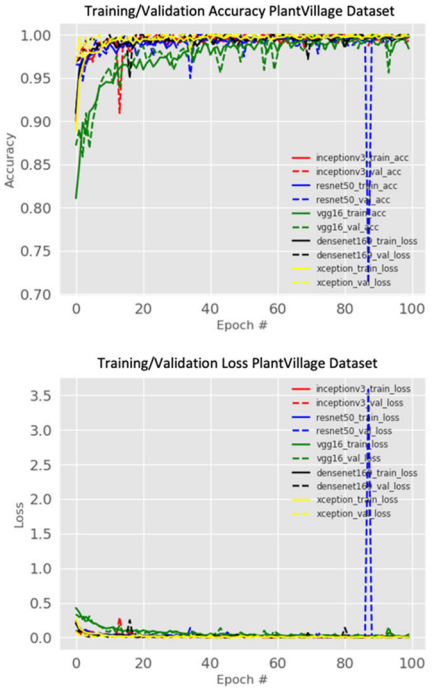


FIGURE 7. Training and validation of DL models using the PlantVillage dataset when NLB and GLS disease images were used.

TABLE 8. Results for PlantVillage dataset with for binary NLB & GLS identification.

Model	NLB	GLS	Total
InceptionV3	97.54	98.91	98.32
ResNet50	97.54	96.56	97.03
VGG16	99.11	96.39	97.56
DenseNet169	98.21	98.07	98.13
Xception	98.88	98.15	**98.47**

All values are in percentages.

and GLS diseases were common to PlantVillage, PlantDoc, and custom CD&S. Therefore, the models were trained to identify NLB and GLS for comparing the generalization testing accuracies across datasets.

After training the models, it was observed that testing accuracies were consistently greater than 97%, with the highest testing accuracy of 98.47% for the Xception model, as shown in table 8. Training and validation were also plotted (figure 7).

2) PLANTDOC DATASET

The same experiment was conducted using the NLB and GLS images from the PlantDoc dataset. Using a smaller number of field images resulted in a much higher testing accuracy across datasets than when training the models using the larger PlantVillage dataset with uniform background. The highest testing accuracy of 88.24% was observed with the Xception model, as shown in table 9. It was also observed that the VGG16 model performed the worst as it could not learn any NLB features resulting in a 100% identification rate for GLS. The plots in figure 8 show the training and validation accuracy and loss.

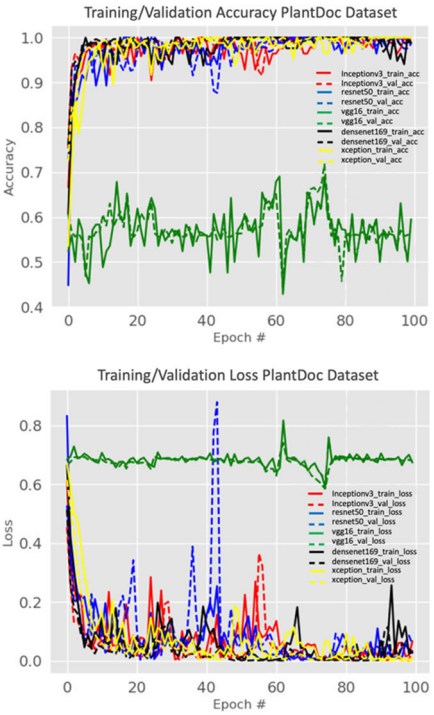


FIGURE 8. Training and validation of DL models using the PlantDoc dataset when NLB and GLS disease images were used.

3) CD&S DATASET

Finally, the NLB and GLS images from the Custom CD&S dataset were used for training the models. The training and validation results can be seen in figure 9. Testing was then performed on images not used for training (table 10). The

TABLE 9. Results for PlantDoc dataset with for binary NLB & GLS identification.

Model	NLB	GLS	Total
InceptionV3	100	74.14	85.29
ResNet50	72.73	96.55	86.27
VGG16	0	100	56.87
DenseNet169	86.36	87.93	87.25
Xception	90.91	86.20	**88.24**

All values are in percentages.

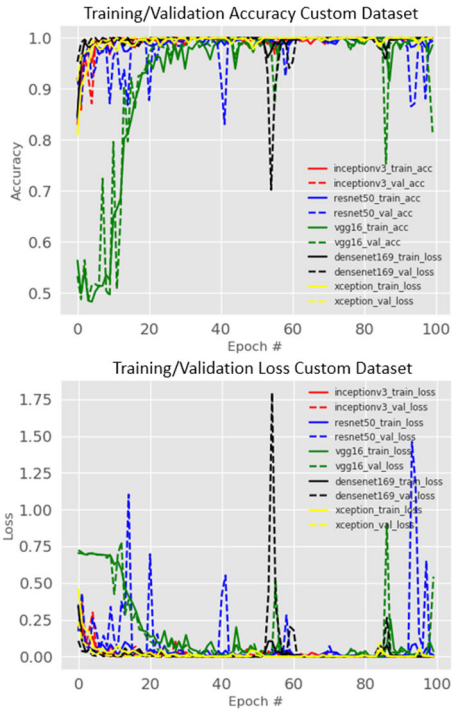


FIGURE 9. Training and validation of DL models using the custom CD&S dataset when NLB and GLS disease images were used.

TABLE 10. Results for CD&S dataset with for binary NLB & GLS identification.

Model	NLB	GLS	Total
InceptionV3	100	99.19	99.61
ResNet50	97.70	99.61	98.43
VGG16	91.19	100	95.48
DenseNet169	100	99.60	**99.80**
Xception	99.23	99.61	99.41

All values are in percentages.

highest testing accuracy was observed for the DenseNet169 model at 99.80%. In addition, the CD&S dataset also consists of field images, and the lowest testing accuracy was again observed for the VGG16 model.

C. COMBINING UNIFORM & FIELD CONDITION IMAGES

1) PLANTVILLAGE & PLANTDOC DATASET

After achieving high testing accuracies across datasets, models trained solely on the PlantVillage dataset were observed to have difficulty identifying diseases from field conditions. Therefore, the PlantVillage dataset was combined with the PlantDoc dataset to obtain a dataset with mixed images from

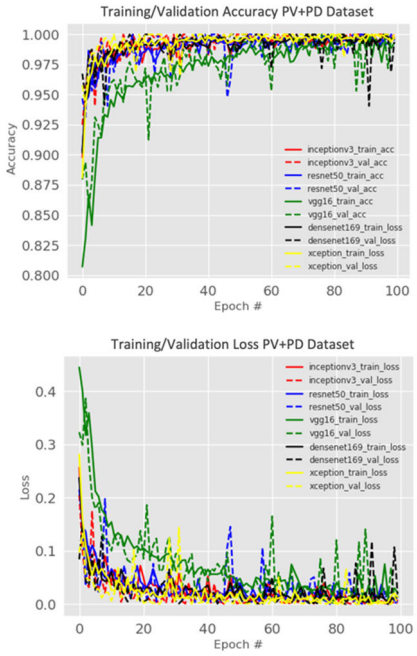


FIGURE 10. Training and validation of DL models using the combined PlantVillage and PlantDoc datasets with NLB and GLS.

TABLE 11. Results for combined PlantVillage & PlantDoc dataset.

	PlantVillage		PlantDoc		
Model	NLB	GLS	NLB	GLS	Total
InceptionV3	95	99	76	95	91.25
ResNet50	97	98	67	98	90.00
VGG16	98	99	62	91	87.50
DenseNet169	95	98	84	84	90.25
Xception	99	97	90	93	**94.75**

All values are in percentages.

uniform and field conditions. The same five models were trained using the same hyperparameters to maintain consistency (figure 10).

Once training was completed, the Xception model achieved the highest testing accuracy of 94.75% when the PlantVillage and PlantDoc testing datasets were used, as shown in table 11.

2) PLANTVILLAGE AND CD&S DATASET

However, as the PlantDoc dataset was not well organized and contained different types of images acquired from different sources, the same test was conducted by replacing the PlantDoc dataset with the CD&S dataset acquired in field conditions (figure 11). After training the five pre-trained models on the combined PlantVillage and CD&S testing datasets, high testing accuracies were observed, with the DenseNet169 model achieving the highest testing accuracy of 98.75% (table 12).

3) PLANTVILLAGE, PLANTDOC, & CD&S DATASET

A final experiment was conducted by combining the PlantVillage, PlantDoc, and CD&S datasets to improve the generalization and testing accuracy across datasets. The Xception

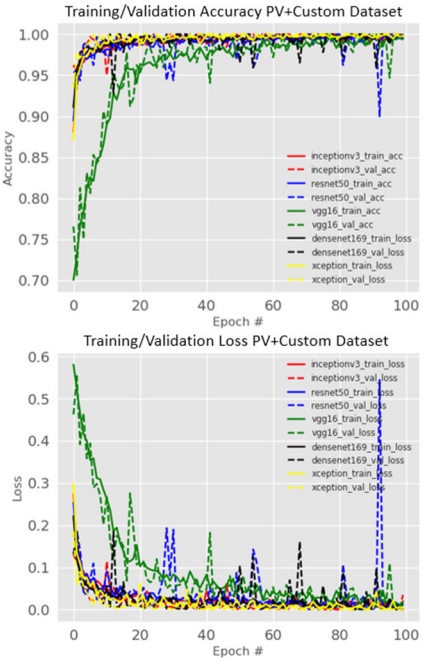


FIGURE 11. Training and validation of DL models using the combined PlantVillage and CD&S datasets with NLB and GLS.

TABLE 12. Results for combined PlantVillage & CD&s dataset.

	PlantVillage		CD&S		
Model	NLB	GLS	NLB	GLS	Total
InceptionV3	98	96	98	100	98.00
ResNet50	98	99	86	100	98.25
VGG16	99	96	99	99	98.25
DenseNet169	99	97	100	99	**98.75**
Xception	98	98	100	98	98.50

All values are in percentages.

TABLE 13. Results for combined PlantVillage, PlantDoc, & CD&S dataset.

	PlantVillage		PlantDoc		CD&S		
Model	NLB	GLS	NLB	GLS	NLB	GLS	Total
InceptionV3	98	99	91	84	100	100	95.33
ResNet50	88	100	79	95	100	98	93.33
VGG16	97	97	67	93	96	98	91.33
DenseNet169	96	99	93	89	100	97	95.67
Xception	98	99	86	93	99	100	**95.83**

All values are in percentages.

model obtained the highest testing accuracy of 95.83%, as shown in table 13.

D. GENERALIZATION ACROSS DATASETS

1) PLANTVILLAGE DATASET

The results for generalization accuracies across datasets are shown in table 14. First, the five models trained using the PlantVillage dataset were provided with testing data from all other datasets used in this study. As the PlantVillage dataset contains the NLB, GLS, common rust, and healthy images for corn, the models could identify these classes. However, the NLS images from the Custom CD&S dataset were not provided when providing testing data as the model was not trained for the NLS disease. After providing testing data from different datasets, the testing accuracies for InceptionV3, ResNet50, VGG16, DenseNet169, and Xception were 34.82%, 32.45%, 43.09%, 37.64%, and 45.36%.

Initial testing using the PlantVillage dataset resulted in the DenseNet169 model achieving the highest testing accuracy of 99.50%. However, the highest testing accuracy during the generalization test of 45.36% was achieved with the Xception model. As the models trained using the PlantVillage dataset were trained on images acquired in controlled lab conditions, it was observed the models had difficulty generalizing to the data acquired in field conditions.

The observations for DL model generalization using the PlantVillage dataset confirms the findings reported in the literature [6], [26], [27], [28]. Interestingly, when the custom CD&S dataset with a white background was used for testing, the DenseNet169 model could identify the NLB and GLS images with accuracies of around 55% and 76%, respectively, which was the highest overall testing accuracy for a particular dataset. However, when the black background images were used for testing, no model could accurately identify the NLB disease.

2) PLANTDOC DATASET

The models trained using the PlantDoc dataset from the first experiment were then used to run the generalization tests. During the testing for PlantDoc, the images from PlantDoc were disregarded, and the PlantVillage images for NLB, GLS, and common rust were used. Once again, the NLS images were also disregarded as the models were not trained to identify them. After providing the models with the testing images, the accuracy achieved with InceptionV3, ResNet50, VGG16, DenseNet169, and Xception was 58.55%, 52.36%, 52.55%, 61.00%, and 58.00%, respectively. This time, the highest testing accuracy was achieved for the DenseNet169 model. During this experiment, it was observed that the trained models had the greatest difficulty in identifying the GLS images from across datasets, with ResNet50 performing the best for identifying GLS.

The initial highest testing accuracy when testing images from PlantDoc were provided was 81.53% for the Xception model. Although the highest testing accuracy dropped to 61.00% during the generalization tests, it is still 15.64% higher than the highest observed testing accuracy for the generalization experiments for the PlantVillage-trained models, which was 45.36%.

3) CD&S DATASET

After reporting the generalization performance of the models trained using publicly available corn disease datasets, the performance of models trained using the CD&S dataset was also evaluated. The testing images from the common rust and rust classes in PlantVillage and PlantDoc were not used. All other classes were used, including the CD&S datasets

TABLE 14. Results for generalization across datasets.

			PlantVillage			PlantDoc			Digipathos	NLB Dataset	CD&S			CD&S White Background			CD&S Black Background			
		Model	NLB	GLS	Rust	NLB	GLS	Rust	NLB	NLB	NLB	GLS	NLS	NLB	GLS	NLS	NLB	GLS	NLS	Total
Entire Datasets Used for Training	PlantVillage	InceptionV3	-	-	-	17	91	22	33	15	3	94	-	37	66	-	0	5	-	34.82%
		ResNet50	-	-	-	22	86	16	23	9	4	98	-	10	85	-	0	4	-	32.45%
		VGG16	-	-	-	40	89	9	41	55	6	99	-	47	76	-	0	12	-	43.09%
		DenseNet169	-	-	-	36	77	13	41	27	5	82	-	55	76	-	0	2	-	37.64%
		Xception	-	-	-	38	89	4	42	25	7	99	-	36	95	-	0	64	-	**45.36%**
	PlantDoc	InceptionV3	72	17	83	-	-	-	68	85	96	11	-	96	10	-	90	16	-	58.55%
		ResNet50	49	34	68	-	-	-	44	28	95	32	-	47	65	-	83	31	-	52.36%
		VGG16	56	8	97	-	-	-	48	66	98	5	-	95	3	-	78	24	-	52.55%
		DenseNet169	69	16	90	-	-	-	83	89	100	5	-	99	10	-	98	12	-	**61.00%**
		Xception	69	10	94	-	-	-	61	89	92	20	-	82	36	-	90	13	-	58.00%
	CD&S	InceptionV3	8	70	-	60	52	-	33	44	-	-	-	0	0	100	43	89	66	47.08%
		ResNet50	32	85	-	56	54	-	44	56	-	-	-	60	75	99	45	100	14	60.00%
		VGG16	3	81	-	65	54	-	56	93	-	-	-	62	68	98	71	96	55	**66.83%**
		DenseNet169	3	71	-	58	52	-	29	75	-	-	-	5	28	100	36	73	97	52.25%
		Xception	37	71	-	68	80	-	47	90	-	-	-	27	98	33	4	100	0	54.58%
	CD&S Back	InceptionV3	24	97	-	49	78	-	30	50	95	93	40	90	94	84	95	96	83	73.27%
		ResNet50	17	97	-	51	63	-	53	87	96	44	76	57	41	100	92	94	91	70.60%
		VGG16	0	0	-	0	0	-	0	0	0	0	100	0	0	100	0	0	100	20.00%
		DenseNet169	22	94	-	67	57	-	71	98	96	87	72	95	96	88	96	96	89	**81.60%**
		Xception	40	97	-	67	72	-	65	98	99	74	37	99	89	71	100	89	85	78.80%
Binary NLB & GLS Datasets Used for Training	PlantVillage	InceptionV3	-	-	-	36	93	-	32	41	2	99	-	30	82	-	2	99	-	51.60%
		ResNet50	-	-	-	34	93	-	44	52	0	100	-	43	82	-	0	100	-	54.80%
		VGG16	-	-	-	40	93	-	44	66	1	100	-	39	79	-	0	100	-	**56.20%**
		DenseNet169	-	-	-	28	100	-	41	36	0	100	-	69	76	-	0	100	-	55.00%
		Xception	-	-	-	29	98	-	35	22	2	100	-	3	100	-	0	100	-	48.90%
	PlantDoc	InceptionV3	11	98	-	-	-	-	47	83	56	75	-	3	100	-	17	98	-	58.80%
		ResNet50	82	53	-	-	-	-	82	95	84	52	-	30	85	-	64	59	-	68.60%
		VGG16	100	0	-	-	-	-	100	100	100	0	-	100	0	-	100	0	-	60.00%
		DenseNet169	41	88	-	-	-	-	83	99	94	54	-	58	85	-	40	91	-	73.30%
		Xception	55	75	-	-	-	-	76	94	89	85	-	72	80	-	94	62	-	**78.20%**
	CD&S	InceptionV3	72	17	-	96	11	-	68	85	-	-	-	51	100	-	1	100	-	60.10%
		ResNet50	49	34	-	95	32	-	44	28	-	-	-	20	98	-	0	100	-	50.00%
		VGG16	56	8	-	98	5	-	48	66	-	-	-	14	79	-	0	100	-	47.40%
		DenseNet169	69	16	-	100	5	-	83	89	-	-	-	55	94	-	43	97	-	**65.10%**
		Xception	69	10	-	92	20	-	61	89	-	-	-	10	100	-	1	100	-	55.20%
Binary NLB & GLS Combined Datasets Used	PV + PD	InceptionV3	-	-	-	-	-	-	74	91	77	84	-	51	100	-	1	100	-	72.25%
		ResNet50	-	-	-	-	-	-	80	77	77	88	-	20	98	-	0	100	-	67.50%
		VGG16	-	-	-	-	-	-	73	88	83	37	-	14	79	-	0	100	-	59.25%
		DenseNet169	-	-	-	-	-	-	86	90	95	76	-	55	94	-	43	97	-	**79.50%**
		Xception	-	-	-	-	-	-	71	99	86	81	-	10	100	-	1	100	-	68.50%
	PV + CD&S	InceptionV3	-	-	-	60	89	-	75	100	-	-	-	97	71	-	1	100	-	74.13%
		ResNet50	-	-	-	31	98	-	45	95	-	-	-	38	100	-	0	100	-	63.38%
		VGG16	-	-	-	60	91	-	52	95	-	-	-	99	75	-	0	100	-	71.50%
		DenseNet169	-	-	-	79	86	-	67	99	-	-	-	100	76	-	13	100	-	**77.50%**
		Xception	-	-	-	79	86	-	79	99	-	-	-	54	100	-	14	100	-	76.38%
	PV+PD+CD&S	InceptionV3	-	-	-	-	-	-	71	100	-	-	-	94	95	-	21	100	-	80.17%
		ResNet50	-	-	-	-	-	-	77	90	-	-	-	38	100	-	0	100	-	67.50%
		VGG16	-	-	-	-	-	-	80	94	-	-	-	67	98	-	45	98	-	**80.33%**
		DenseNet169	-	-	-	-	-	-	82	93	-	-	-	71	89	-	0	100	-	72.50%
		Xception	-	-	-	-	-	-	71	82	-	-	-	57	100	-	17	100	-	71.17%

All values are in percentages.

with white and black backgrounds. The testing accuracies for InceptionV3, ResNet50, VGG16, DenseNet169, and Xception of 47.08%, 60.00%, 66.83%, 52.25%, and 54.58% were achieved, respectively. Although the VGG16 model performed the best with a testing accuracy of 66.83%, the Xception and ResNet50 models were superior when identifying the NLB and GLS images from PlantVillage and PlantDoc datasets.

4) CD&S DATASET WITH REMOVED BACKGROUND

All experiments conducted for this study consisted of either uniform backgrounds or field conditions with noisy backgrounds. When the models trained on such datasets were then used for evaluating generalization, the testing accuracies dropped significantly.

The generalization performance of models trained using the CD&S dataset with the removed background was then provided with testing images from all the datasets used in this study. Once again, the common rust and rust images from the PlantVillage and PlantDoc datasets were removed. The testing accuracies for InceptionV3, ResNet50, VGG16, DenseNet169, and Xception of 73.27%, 70.60%, 20.00%, 81.60%, and 78.80% were achieved, respectively. The lowest accuracy was observed for VGG16, which failed to

train well as it consistently identified all images as NLS. The DenseNet169 model once again achieved the highest accuracy. However, the best performance for identifying the images from PlantVillage and PlantDoc datasets was achieved by the Xception model.

The testing accuracy for generalizing to different datasets and conditions of 81.60% achieved by the DenseNet169 model trained using the CD&S dataset with no background was the highest testing accuracy achieved during the generalization tests.

5) PLANTVILLAGE DATASET BINARY NLB & GLS

Once the generalization performance for the models trained on entire corn disease datasets was reported, the generalization tests were performed for the models trained to identify only the NLB and GLS diseases of corn using the PlantVillage dataset. During testing, only NLB and GLS images were provided for testing from all the datasets except for PlantVillage. The testing accuracies for InceptionV3, ResNet50, VGG16, DenseNet169, and Xception of 51.60%, 54.80%, 56.20%, 55.00%, and 48.90% were achieved, respectively. VGG16 achieved the highest testing accuracy at 56.20%, which was higher than the 45.36% testing accuracy achieved during the generalization performance evaluation when the entire PlantVillage dataset was used.

6) PLANTDOC DATASET BINARY NLB & GLS

The generalization performance was also evaluated for the models trained using only NLB and GLS images from the PlantDoc dataset. The testing accuracies for InceptionV3, ResNet50, VGG16, DenseNet169, and Xception of 58.80%, 68.60%, 60.00%, 73.30%, and 78.20% were achieved, respectively. The highest testing accuracy of 78.20% achieved by the Xception model was 17.20% higher than when the entire PlantDoc dataset was used for training.

7) CD&S DATASET BINARY NLB & GLS

The same experiment was conducted for the Custom CD&S dataset by training the models using only NLB and GLS images. The testing accuracies achieved for evaluating the generalization performance for InceptionV3, ResNet50, VGG16, DenseNet169, and Xception of 60.10%, 50.00%, 47.40%, 65.10%, and 55.20% were achieved, respectively. Surprisingly, in this case, the best testing accuracy of 65.10% achieved by the DenseNet169 model was slightly lower than the 66.83% achieved when the entire dataset, including the NLS images, was used to train the models.

8) PLANTVILLAGE & PLANTDOC DATASET

The results indicate that by adding 10% of images acquired in field conditions to a dataset of images acquired in controlled lab conditions, the testing accuracy across datasets increased from 45.36% to 79.50% for the PlantVillage dataset and from 61.00% to 79.50% for the PlantDoc dataset. The highest testing accuracy on the same dataset used for training the models was 94.75%. However, the overall testing accuracy across datasets not used for training was 79.50% with the DenseNet169 model.

9) PLANTVILLAGE & CD&S DATASET

When the PlantVillage and CD&S datasets were combined, testing accuracies up to 98.75% were achieved when training and testing images from the same dataset were used. However, the overall testing accuracies across datasets was 77.50% which was 2% lower than when the CD&S dataset replaced the PlantDoc dataset. This was due to the lower testing accuracies when identifying the NLB testing images from the CD&S dataset with a black background.

10) PLANTVILLAGE, PLANTDOC, & CD&S DATASET

Although the testing accuracy on the PlantVillage, PlantDoc, and CD&S testing datasets reached up to 95.83%, the VGG16 model achieved the highest testing accuracy across datasets at 80.33%. Therefore, it was clear that by adding a small percentage of data consisting of images acquired in field conditions to a dataset of images acquired in controlled lab conditions, the generalization performance of such models improved. Combining multiple different dataset types can be a promising approach for improving DL model generalization for disease identification.

IV. DISCUSSION

State-of-the-art computer vision models rely on DL for identifying and diagnosing plant diseases using imagery data. However, disease identification is a complex problem due to the small lesion sizes, similarity in lesion features, and variability in different background conditions. Therefore, it is important for trained DL models to be able to accurately identify diseases despite the differences in external factors. The difficulty of DL models to generalize well across these different conditions for plant disease identification has been identified as a research gap. Recent studies have struggled with training generalizable models. Therefore, this study focused on thoroughly analyzing how well DL models can train using popular datasets. In addition, the study was focused on evaluating different dataset combinations as an augmentation technique. Finally, all trained models were used to evaluate their ability to generalize to hidden testing images from different datasets and image acquisition conditions.

Overall, the most popular publicly available plant disease dataset, PlantVillage, achieved testing accuracies of up to 99.50% when all the corn disease classes were used for training. The CD&S and PlantDoc datasets achieved testing accuracies of up to 81.53% and 99.62%, respectively. For binary NLB and GLS identification, promising performance was observed using PlantVillage, PlantDoc, and CD&S datasets with testing accuracies of up to 98.47%, 88.24%, and 99.80%, respectively. When assessing the generalization performance, the DL models trained using the PlantDoc dataset performed better than when the PlantVillage dataset was used, as shown in table 14. This was due to the complex field background within the PlantDoc dataset as opposed to the

TABLE 15. Testing accuracies for same training and testing datasets.

Study	Dataset	Testing Accuracy
[25]	Lab (PlantVillage) / Lab (PlantVillage)	99.53%
[6], [26]	Lab (PlantVillage) / Lab (PlantVillage)	99.01%
[6]	Field (PlantDoc) / Field (PlantDoc)	70.53%
[27]	Lab (PlantVillage) / Lab (PlantVillage)	99.84%
Our	PlantVillage	99.50%
	PlantDoc	81.53%
	CD&S	99.62%
	CD&S No Background	87.52%
	PlantVillage + PlantDoc	94.75%
	PlantVillage + CD&S	98.75%
	PlantVillage + PlantDoc + CD&S	95.83%

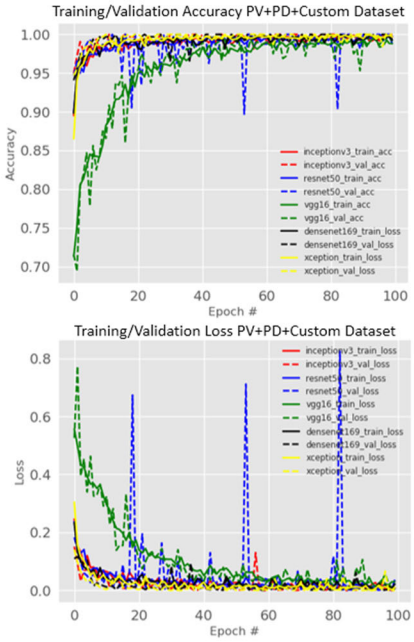


FIGURE 12. Training and validation of DL models using the combined PlantVillage, PlantDoc, and CD&S datasets with NLB and GLS.

uniform backgrounds for the corn disease images within the PlantVillage dataset. The best performances were observed when the background for the CD&S dataset was removed and when datasets with different background conditions were combined.

When training and testing were conducted using the same datasets, the results from this study were comparable to the results in the literature. Testing accuracies of greater than 99% were reported (table 15, figure 13).

For generalization across datasets and image acquisition conditions, this study demonstrated that background removal using RGBA images or augmentation of field data with lab data is a promising approach. Augmentation of different data types resulted in improved performance due to diversity in training data. Reasons for both these outcomes are discussed later. Testing accuracies in this study were up to 10% higher than the overall best performance reported in the literature. The results from this study were compared with results reported in the literature (table 16).

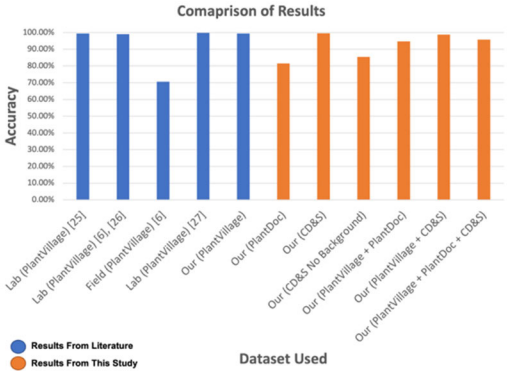


FIGURE 13. Comparison of testing accuracy results reported in the literature with those reported in this study.

TABLE 16. Generalization accuracy across conditions & datasets.

Study	Train / Test (Dataset Type)	Testing Accuracy
[25], [26]	Lab (PlantVillage) / Field	33.27%
	Field / Lab (PlantVillage)	65.69%
[26]	Lab (PlantVillage) / Field (IPM)	45.95%
	Lab (PlantVillage) / Field (Bing)	33.97%
[27]	Lab (PlantVillage) / Field	39.38%
	Lab (PlantVillage) / Field	72.03%
[6]	Lab (PlantVillage) / Field (PlantDoc)	39.87%
Our	Lab (PlantVillage) / Field (PlantDoc)	46.00%
	Lab (PlantVillage) / Field (CD&S)	53.00%
	Lab (PlantVillage) / Field (Digipathos)	42.00%
	Lab (PlantVillage) / Field (NLB Dataset)	55.00%
	Lab (PlantVillage) / Lab (CD&S White)	65.50%
	Lab (PlantVillage) / Lab (CD&S Black)	32.00%
	Field (PlantDoc) / Lab (PlantVillage)	58.33%
	Field (PlantDoc) / Field (CD&S)	63.50%
	Field (PlantDoc) / Field (Digipathos)	83.00%
	Field (PlantDoc) / Field (NLB Dataset)	89.00%
	Field (PlantDoc) / Lab (CD&S White)	59.00%
	Field (PlantDoc) / Lab (CD&S Black)	57.00%
	Field (CD&S) / Field (PlantVillage)	58.50%
	Field (CD&S) / Lab (PlantDoc)	74.00%
	Field (CD&S) / Field (Digipathos)	56.00%
	Field (CD&S) / Field (NLB Dataset)	93.00%
	Field (CD&S) / Lab (CD&S White)	78.00%
	Field (CD&S) / Lab (CD&S Black)	74.00%
	Field (CD&S No BG) / Field (CD&S)	85.00%
	Field (CD&S No BG) / Field (PlantVillage)	68.50%
	Field (CD&S No BG) / Lab (PlantDoc)	69.50%
	Field (CD&S No BG) / Field (Digipathos)	71.00%
	Field (CD&S No BG) / Field (NLB Dataset)	98.00%
	Field (CD&S No BG) / Lab (CD&S White)	93.00%
	Field (CD&S No BG) / Lab (CD&S Black)	93.67%

In general, promising generalization performance was observed in this study when RGB images were converted to RGBA images for background removal as this allowed DL models to learn important features corresponding to disease lesions for distinct disease types rather than being influenced by features pertaining to background variations [38]. High average testing accuracy of 81.60% was achieved for generalization across all datasets, which was higher than the results reported in the literature of 33.27%, 65.69%, 33.97%, 45.95%, and 72.03% [6], [25], [26], [27]. It was also observed

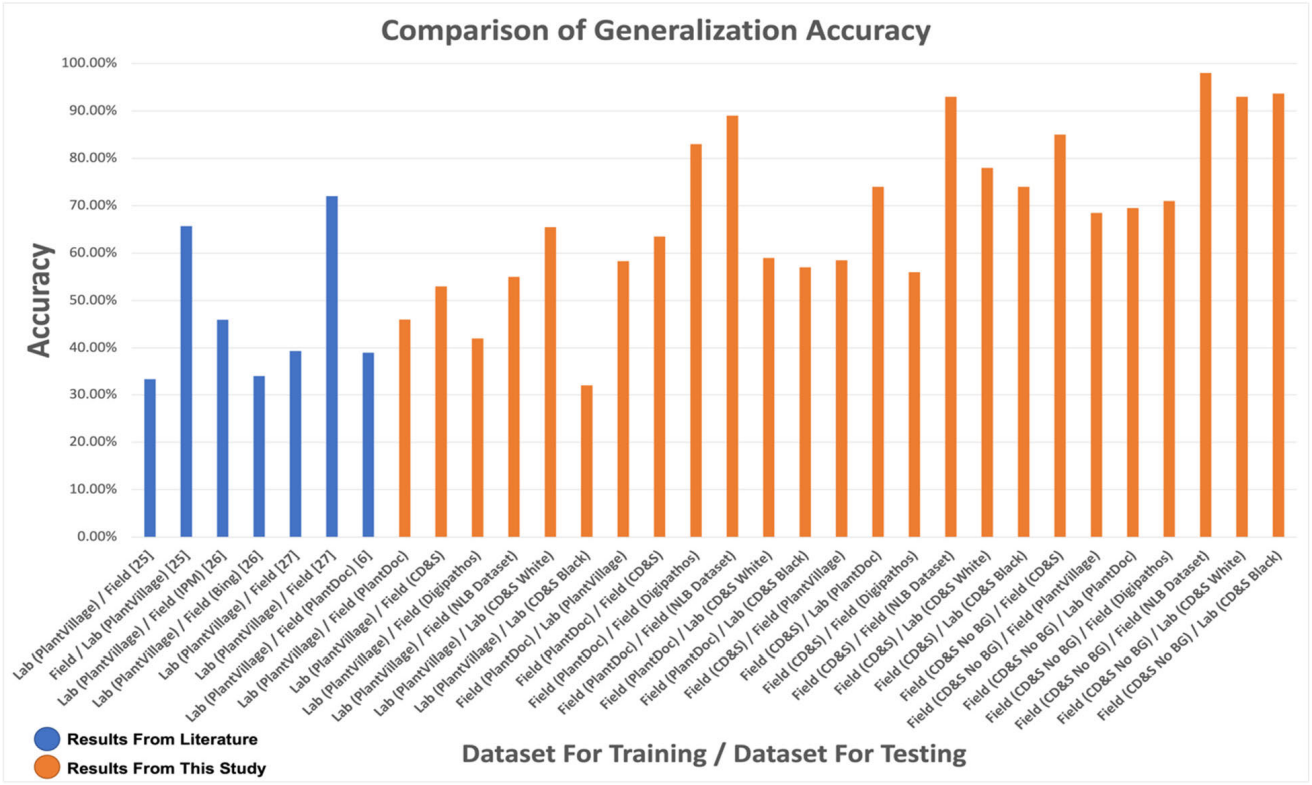


FIGURE 14. Comparison of generalization ability of DL models trained in this study for plant disease identification with results reported in literature.

that the generalization performance from multiple experiments in this study was superior to the results reported in the literature (figure 14).

The performance of the DL models trained using RGBA images was further confirmed using the modified RGBA images with randomized RGB channels within the alpha (A) regions where background was removed. After training the models, their performance remained unchanged, ensuring that the DL models were not influenced by image pixels corresponding to removed background regions. Therefore, the DL models learned the important features corresponding to each class's diseased corn leaves and the disease lesions.

Additionally, DL models were also trained using only NLB and GLS disease classes for training binary image classification models. This experiment was carried out as the NLB and GLS classes were common to the PlantVillage, PlantDoc, and CD&S datasets. Therefore, the models could help obtain a more accurate evaluation of the generalization performance. The overall generalization results obtained when only NLB and GLS classes were identified are shown in table 17. By augmenting the different datasets, generalization accuracy under field conditions increased to 85.50% when multiple different plant disease classes from CD&S dataset were used for testing and 100% when a single disease NLB dataset was used for testing (table 17). The results suggest that due to diversity in the training data, DL models were able to learn important disease features under diverse environmental conditions, thereby resulting in improved generalization performance.

TABLE 17. Generalization accuracy across conditions & datasets after combining training datasets.

Train / Test (Dataset Type)	Testing Accuracy
Field & Lab (PlantVillage & PlantDoc) / Field (CD&S)	85.50%
Field & Lab (PlantVillage & PlantDoc) / Field (Digipathos)	86.00%
Field & Lab (PlantVillage & PlantDoc) / Field (NLB Dataset)	99.00%
Field & Lab (PlantVillage & PlantDoc) / Lab (CD&S White)	75.50%
Field & Lab (PlantVillage & PlantDoc) / Lab (CD&S Black)	70.00%
Field & Lab (PlantVillage & CD&S) / Field (PlantDoc)	82.50%
Field & Lab (PlantVillage & CD&S) / Field (Digipathos)	79.00%
Field & Lab (PlantVillage & CD&S) / Field (NLB Dataset)	100%
Field & Lab (PlantVillage & CD&S) / Lab (CD&S White)	88.00%
Field & Lab (PlantVillage & CD&S) / Lab (CD&S Black)	57.00%
Field & Lab (PlantVillage& PlantDoc & CD&S) / Field (Digipathos)	82.00%
Field & Lab (PlantVillage& PlantDoc & CD&S) / Field (NLB Dataset)	100%
Field & Lab (PlantVillage& PlantDoc & CD&S) / Lab (CD&S White)	94.50%
Field & Lab (PlantVillage& PlantDoc & CD&S) / Lab (CD&S Black)	71.50%

V. CONCLUSION

This study evaluated the generalization capability of DL models for identifying plant diseases under diverse environmental conditions. Five pre-trained DNN architectures, namely VGG16, ResNet50, InceptionV3, DenseNet169, and Xception, were used to train and assess DL models by utilizing corn disease images from different datasets. Transfer learning

using DenseNet169 resulted in the best generalization for plant disease identification, capable of achieving testing accuracies up to 81.60%. The best generalization accuracy across datasets and background conditions of 81.60% was observed when the DenseNet169 model was trained on the RGBA images from the CD&S dataset with removed backgrounds. This can be attributed to DenseNet169 DNN ability to retain high level features and the RGBA images with removed backgrounds which likely helped the model focus on relevant disease features. In addition, high generalization accuracies of up to 80.33% were observed when DL models were trained by combining lab and field-acquired images corresponding to diverse environmental conditions as this introduced diversity in training data. Overall, DL model generalization improved due to the use of RGBA images for background removal and augmentation of different data types. Therefore, this study could be used for developing field-deployable disease management systems under diverse environmental conditions.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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